

Note 6.1 Eigenvalues and Eigenvectors

Intro

Definition of Eigenvalue(Characteristic Value) and Eigenvector(Characteristic Vector)

For an $n \times n$ matrix A , if there exist a nonzero vector $\mathbf{x}$ and a scalar λ such that $A\mathbf{x} = \lambda\mathbf{x}$, then the scalar is eigenvalue or characteristic value of A and the vector is an eigenvector or characteristic vector belonging to λ .

And any nonzero multiple of $\mathbf{x}$ will be an eigenvector.

To find λ and $\mathbf{x}$, We rewrite the equation as

$$(A - \lambda I)\mathbf{x} = \mathbf{0}$$

since $\mathbf{x} \neq \mathbf{0}$, the matrix $A - \lambda I$ is singular. Then we can obtain the eigenvalue λ by the determinant.

Then substitute λ to the equation and solve the linear system we can get the eigenvector $\mathbf{x}$.

And the subspace $N(A - \lambda I)$ is called the eigenspace corresponding to eigenvalue λ , the polynomial $p(\lambda) = \det(A - \lambda I)$ is characteristic polynomial and the determinant is characteristic equation.

Equivalence of Eigenvalue

For an $n \times n$ matrix and a scalar λ , the following statement are equivalent.

- λ is an eigenvalue of A .
- $(A - \lambda I)\mathbf{x} = \mathbf{0}$ has a nontrivial solution.
- $N(A - \lambda I) \neq \{\mathbf{0}\}$.
- $A - \lambda I$ is singular.
- $\det(A - \lambda I) = 0$.

Complex Eigenvalues

Let a eigenvalue λ be a eigenvalue of A and $\mathbf{z}$ is the eigenvector belonging to λ , then their conjugate are also have the property. That is, $\bar{\lambda}$ is also a eigenvalue of A and $\bar{\mathbf{z}}$ is an eigenvector belonging to $\bar{\lambda}$.

For the conjugate of matrix, we have $\bar{A} = (\bar{a}_{ij})$ and $\overline{AB} = \bar{A}\bar{B}$.

The Product and Sum of the Eigenvalues

For the characteristic equation of A , we have the following properties.

- The product of eigenvalues equals to the determinant of A . That is,

$$\prod_{i=1}^n \lambda_i = \det(A)$$

- The sum of eigenvalues equals to the sum of the diagonal elements of A (call trace and denoted by $\text{tr}(A)$). That is,

$$\sum_{i=1}^n \lambda_i = \sum_{i=1}^n a_{ii}$$

Similar Matrices

Theorem 6.1.1

For two similar matrices, they have the same characteristic polynomials and the same eigenvalues.